

A Dual-Approach System for Numeral-Aware Reasoning in Natural Language Processing

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ABSTRACT

This report presents a comprehensive study of the Numeral-Aware Reasoning project, which addresses a key limitation in modern natural language processing models: their inability to perform robust and relational numerical reasoning. The project introduces a dual-approach methodology consisting of a symbolic reasoning engine and a transformer-based neural system fine-tuned for numerical inference and masked number prediction. The symbolic system offers deterministic, interpretable reasoning for simple arithmetic event sequences, while the neural system achieves strong empirical performance using regression-based numerical prediction and numerical natural language inference tasks. Experimental results demonstrate that the NLI model achieved 67.0

Introduction

Numerical reasoning remains a persistent challenge in natural language processing. Typical transformer-based large language models treat numbers as isolated lexical tokens rather than structured values with inherent relational meaning. For example, the difference between “7” and “8” is not encoded numerically but lexically, limiting the model’s ability to perform arithmetic and numerical inference. This project seeks to bridge that gap by developing two complementary systems capable of performing numeral-aware reasoning: a symbolic, rule-based reasoning engine and a neural model fine-tuned for numerical inference. Together, these systems provide insight into the strengths and weaknesses of symbolic versus statistical numeric reasoning.

Results

Symbolic Reasoning Engine

The symbolic system parses sentences to extract entities, quantities, items, and arithmetic events such as “had” or “gave away.” Using this information, it constructs a world state and performs deterministic reasoning. The system achieves perfect accuracy within its defined rule set but is limited by linguistic brittleness.

Neural Reasoning Models

Two neural models were developed: a numerical NLI classifier and a masked number regression model. The NLI model reached a final test accuracy of 67.0%, demonstrating its ability to learn numerical-linguistic patterns. The regression model reached a validation MAE of 1.636, highlighting its effectiveness in predicting continuous-valued numerical outputs.

Discussion

The symbolic system excels in interpretability but fails on linguistic variability. Conversely, the neural models generalize well across text patterns but do not perform explicit arithmetic reasoning. Their numerical abilities emerge statistically rather than procedurally. These trade-offs motivate future work in neuro-symbolic integration, wherein symbolic world states could be used alongside neural contextual embeddings to improve numerical reasoning robustness.

Methods

The symbolic engine was implemented using Python with rule-based parsing and dictionary-based world state tracking. Neural models were implemented in PyTorch using the Hugging Face transformers library. DistilBERT served as the encoder for both NLI and regression tasks. Training used standard benchmarks such as NumGLUE, EQUATE, and synthetic datasets.

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Author contributions statement

Sagnik Paira designed the system architecture, implemented the symbolic engine, and trained the neural models. Previous reports assisted in structuring, refining, and documenting the report. All authors reviewed the manuscript.

Additional information

The authors declare no competing interests.